

ICT Class 10 - Formative Assessment 1

Formative Assessment (FA-1)

- Class 10

Class: Class 10

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Networking Concepts, Internet and Web Technologies

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which of the following is NOT a type of computer network?
 - a. LAN
 - b. WAN
 - c. MAN
 - d. CAN
- b. The full form of TCP/IP is:
 - a. Transfer Control Protocol/Internet Protocol
 - b. Transmission Control Protocol/Internet Protocol
 - c. Transmission Communication Protocol/Internet Protocol
 - d. Transfer Communication Protocol/Internetwork Protocol
- c. Which device is used to connect multiple networks and direct data packets?
 - a. Hub
 - b. Switch
 - c. Router
 - d. Modem
- d. The top-level domain .edu is used for:
 - a. Government websites
 - b. Educational institutions
 - c. Commercial organisations
 - d. Network infrastructure
- e. What is the maximum number of characters allowed in a URL as per standard practice?
 - a. 2048
 - b. 255
 - c. 1024
 - d. 512

Section II: Short Answer (5 x 2 = 10 marks)

- a. Differentiate between **LAN** and **WAN** with two examples each.

- Explain the client-server model with a neat labelled diagram.
- What is the role of a DNS server? Explain how domain name resolution works.
- Compare HTTP and HTTPS. Why is HTTPS preferred for banking websites?
- What is cloud computing? List any two service models (IaaS, PaaS, SaaS).

Section III: Long Answer (5 x 1 = 5 marks)

- Explain the TCP/IP protocol suite with the help of a diagram. Describe the function of each layer and give one protocol used at each layer. **(Board Exam Pattern - 2024)**

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Reads a CSV file containing student data (Name, Roll, Marks in 3 subjects)
- Calculates total and percentage for each student
- Writes the results to a new file with Grade column (≥ 90 : A, ≥ 75 : B, ≥ 60 : C, < 60 : D)
- Displays the topper of the class

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently

Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions
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Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 10 - Formative Assessment 2

Formative Assessment (FA-2)

- Class 10

Class: Class 10

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: HTML Fundamentals, CSS Styling

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General Instructions

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3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which HTML tag is used to create an ordered list?
 - b. `<ul>`
 - c. `<ol>`
 - d. `<li>`
 - e. `<list>`
- b. Which CSS property is used to change the text colour?
 - c. `font-color`
 - d. `text-color`
 - e. `color`
 - f. `foreground`
- c. The correct CSS syntax to select an element with id `main` is:
 - d. `.main`
 - e. `#main`
 - f. `main`
 - g. `*main`
- d. Which HTML attribute is used to specify an image source?
 - e. `src`
 - f. `href`
 - g. `link`
 - h. `source`
- e. In CSS, which property controls the space between the element border and its content?
 - f. `margin`
 - g. `padding`
 - `spacing`
 - a. `gap`

Section II: Short Answer (5 x 2 = 10 marks)

- Write the HTML code to create a table with 3 columns (Name, Subject, Marks) and 4 rows of data. Include proper headers.
- Differentiate between **block-level** and **inline** elements with two examples each.
- What is the 盒模型 (box model) in CSS? Explain with a diagram.
- Write the CSS to style a navigation bar with:
 - Horizontal layout using flexbox
 - Background colour dark blue
 - White text links with hover effect
- Explain the difference between **class selector** and **id selector**. When would you use each?

Section III: Long Answer (5 x 1 = 5 marks)

- Create a complete HTML page for a **School Library Management System** homepage that includes:
 - A header with school name and navigation menu
 - A form to search books (title, author, category dropdown)
 - A table displaying 5 books with columns: Title, Author, Category, Availability
 - Proper CSS styling with colours, borders, and responsive layout
 - Footer with copyright information

Previous Year Pattern - Karnataka Board 2023

Part B: Hands-on Practical Lab Task (5 marks)

Write HTML and CSS code for a **Personal Portfolio Website** that includes:

- Hero section with name, designation, and profile image placeholder
- About section with a brief bio
- Skills section using progress bars (HTML/CSS only)
- Contact form with name, email, and message fields
- Responsive design using media queries

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
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Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 10 - Formative Assessment 3

Formative Assessment (FA-3)

- Class 10

Class: Class 10

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Python Programming - Part 1

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General Instructions

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4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which of the following is the correct way to import a module in Python?
 - b. `import module`
 - c. `include module`
 - d. `require module`
 - e. `using module`
- b. What will be the output of `print(type([]))`?
 - c. `<class 'tuple'>`
 - d. `<class 'list'>`
 - e. `<class 'array'>`
 - f. `<class 'dict'>`
- c. Which method is used to read a line from a file in Python?
 - d. `readline()`
 - e. `getLine()`
 - f. `fetchLine()`
 - g. `line()`
- d. What does `with open('file.txt') as f:` do?
 - e. Opens file and closes it automatically after the block
 - f. Opens file and keeps it open forever
 - g. Creates a new file
 - h. Deletes the file
- e. Which exception is raised when dividing by zero in Python?
 - f. `ValueError`
 - g. `ZeroDivisionError`
 - h. `ArithmeticError`
 - i. `RuntimeError`

Section II: Short Answer (5 x 2 = 10 marks)

- a. Write the output of the following code and explain:

```
data = [10, 20, 30, 40, 50]
result = list(filter(lambda x: x > 25, data))
print(result)
```

- Differentiate between `read()`, `readline()`, and `readlines()` file methods.
- Write a Python function that accepts a string and returns a dictionary with character frequency. Example: "hello" -> {'h': 1, 'e': 1, 'l': 2, 'o': 1}
- What is the purpose of `try-except` block? Write an example where it is used for file handling.
- Explain the difference between **syntax error** and **runtime error** with one example each.

Section III: Long Answer (5 x 1 = 5 marks)

- Write a Python program to manage student records using file handling:
 - Store name, roll number, and marks of 5 students in a text file
 - Read the file and display all students sorted by marks (descending)
 - Find and display the student with highest marks
 - Count how many students scored above 75%

Board Exam Pattern - Karnataka 2024

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Reads a text file and counts the number of words, lines, and characters
- Ignores extra spaces and counts unique words
- Displays the 5 most frequently used words
- Writes the summary to a new file

Part C: Notebook Maintenance (5 marks)

- Neatness and organisation of notebook (2 marks)
- Completion of all assigned classwork and homework (2 marks)
- Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 10 - Formative Assessment 4

Formative Assessment (FA-4)

- Class 10

Class: Class 10

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL Database Management

Curated and developed by

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General Instructions

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4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which SQL command is used to retrieve data from a database?
 - a. GET
 - b. RETRIEVE
 - c. SELECT
 - d. FETCH
- b. The WHERE clause in SQL is used with:
 - a. SELECT only
 - b. SELECT, UPDATE, DELETE
 - c. INSERT only
 - d. CREATE only
- c. Which aggregate function returns the total sum of a numeric column?
 - a. COUNT()
 - b. SUM()
 - c. TOTAL()
 - d. ADD()
- d. Which SQL command is used to remove a table completely from the database?
 - a. DELETE TABLE
 - b. REMOVE TABLE
 - c. DROP TABLE
 - d. TRUNCATE TABLE
- e. The GROUP BY clause is used to:
 - a. Sort data
 - b. Filter rows
 - c. Group rows with same values for aggregate functions
 - d. Delete duplicates

Section II: Short Answer (5 x 2 = 10 marks)

- a. Write SQL queries for a table Students(roll_no, name, class, section, marks):

- Find all students of class '10' with marks above 80
 - Count the number of students in each section
 - Find the maximum and minimum marks
 - List students in alphabetical order of names
- b. Differentiate between WHERE and HAVING clause with examples.
- c. What is a **primary key**? Can a table have more than one primary key? Explain.
- d. Write the output of:
- ```
SELECT UPPER('hello world');
SELECT LENGTH('Karnataka');
SELECT CONCAT('ICT', ' Class', ' 10');
```
- e. Explain the ACID properties of a transaction with one example each.

### Section III: Long Answer (5 x 1 = 5 marks)

- a. Write SQL queries for a **Library Management System** with tables:

- Books(book\_id, title, author, price, quantity)
- Members(member\_id, name, phone, membership\_date)
- Issue(issue\_id, book\_id, member\_id, issue\_date, return\_date)

(a) List all books with price above 500 sorted by title (b) Find members who have not returned any book (c) Count how many times each book has been issued (d) Find the total revenue from all books (e) Display members who joined in the last 6 months

**Board Exam Pattern - Karnataka 2023**

## Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to a SQLite database
- Creates tables for a Student Management System
- Inserts at least 10 records
- Performs all CRUD operations (Create, Read, Update, Delete)
- Generates a report with aggregate functions (COUNT, AVG, MAX, MIN)
- Handles database exceptions properly

## Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)

## Part D: Computational Thinking Skills (Internal Assessment)

**Instructions:** Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

| CT Skill                   | Emerging (1)                                               | Developing (2)                                           | Proficient (3)                                                       |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| <b>Decomposition</b>       | Needs help breaking down complex tasks into smaller parts  | Can break down tasks with some guidance or prompts       | Independently breaks down complex problems into manageable sub-tasks |
| <b>Pattern Recognition</b> | Cannot identify similarities or trends in data or problems | Identifies patterns with prompts or scaffolding          | Independently identifies and applies patterns to solve new problems  |
| <b>Abstraction</b>         | Includes all details, cannot filter irrelevant information | Removes some irrelevant details with guidance            | Filters to essential information, focuses on what matters            |
| <b>Algorithm Design</b>    | Provides random or disordered steps                        | Creates sequential but incomplete step-by-step solutions | Designs clear, ordered, complete algorithms with start and end       |
| <b>Debugging</b>           | Gives up when errors occur                                 | Finds and fixes errors with help from teacher or peers   | Systematically tests, identifies, and fixes errors independently     |
| <b>Evaluation</b>          | Does not check if solution works                           | Tests solution with guidance                             | Evaluates efficiency and correctness of own and others' solutions    |

**Student Name: *Date:***

**Total CT Score: / 18 *Teacher Remarks:***

## Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

# ICT Class 10 - Practical Lab Exam 1

## Practical Lab Exam 1

- Class 10

Class: Class 10

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: HTML/CSS and Python Programming

Curated and developed by

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## General Instructions

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4. Marks are indicated against each question.

## Lab Tasks (15 marks)

### Task 1: HTML/CSS - Personal Portfolio Website (5 marks)

Create a complete HTML/CSS webpage for a **Personal Portfolio** that includes:

- Hero section with name, photo placeholder, and tagline
- About section with bio (minimum 3 sentences)
- Skills section displaying at least 5 skills with visual progress bars (HTML/CSS only)
- Projects section with at least 3 project cards (image placeholder, title, description)
- Contact form with name, email, subject, and message fields
- Responsive design using media queries (mobile-friendly)
- Professional styling with consistent colour scheme

### Task 2: Python - Student Records Manager (5 marks)

Write a Python program that:

- Creates a list of dictionaries to store student records (name, roll, marks in 3 subjects)
- Implements a menu-driven interface with options:
  1. Add new student
  2. Display all students
  3. Search student by roll number
  4. Display students sorted by total marks
  5. Display class statistics (average, highest, lowest)
  6. Exit
- Uses proper functions for each operation
- Includes input validation
- Displays results in a formatted table

### Task 3: Python - File Handling with Exception (5 marks)

Write a Python program that:

- Reads a text file (user-specified filename)
- Counts total words, lines, characters, and unique words
- Finds the 5 most frequently used words
- Writes a summary report to a new file
- Uses try-except blocks for:
  - FileNotFoundError

- PermissionError
- General Exception
- Displays proper error messages for each exception type

## Viva Voce (5 marks)

1. What is the difference between `display: none` and `visibility: hidden` in CSS? When would you use each? (1 mark)
2. Explain the 盒模型 (box model) in CSS. What happens when you set `box-sizing: border-box`? (1 mark)
3. Why is `with open()` preferred over `open()` and `close()` in Python file handling? (1 mark)
4. What is the difference between `read()`, `readline()`, and `readlines()` methods? When would you use each? (1 mark)
5. How does Python handle exceptions? What is the purpose of `else` and `finally` blocks in exception handling? (1 mark)

## Part D: Computational Thinking Skills (Internal Assessment)

**Instructions:** Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

| CT Skill                   | Emerging (1)                                               | Developing (2)                                           | Proficient (3)                                                       |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
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| <b>Debugging</b>           | Gives up when errors occur                                 | Finds and fixes errors with help from teacher or peers   | Systematically tests, identifies, and fixes errors independently     |
| <b>Evaluation</b>          | Does not check if solution works                           | Tests solution with guidance                             | Evaluates efficiency and correctness of own and others' solutions    |

**Student Name:** *Date:*

**Total CT Score:** / 18 **Teacher Remarks:**

## Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

# ICT Class 10 - Practical Lab Exam 2

## Practical Lab Exam 2

- Class 10

Class: Class 10

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL Database and Python with Database

Curated and developed by

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## General Instructions

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4. Marks are indicated against each question.

## Lab Tasks (15 marks)

### Task 1: SQL Queries (5 marks)

Write SQL queries for a **Bookstore Database** with the following tables:

- Books(book\_id INT PRIMARY KEY, title VARCHAR(50), author VARCHAR(30), price DECIMAL, genre VARCHAR(20), stock INT)
- Customers(cust\_id INT PRIMARY KEY, name VARCHAR(30), email VARCHAR(50), city VARCHAR(20))
- Sales(sale\_id INT PRIMARY KEY, book\_id INT, cust\_id INT, quantity INT, sale\_date DATE)

Write queries for: (a) List all books with price above 500 sorted by title (b) Find customers from 'Bangalore' or 'Mysore' (c) Count the number of books in each genre (d) Find the total revenue (price — quantity) for each book (e) Find the top 3 selling books by total quantity sold (f) Find customers who have not made any purchase (g) Display the monthly sales summary (month, total\_sales, total\_books\_sold)

### Task 2: Python with SQLite Database (5 marks)

Write a Python program that:

- Creates a SQLite database for a **Student Result System**
- Creates tables: Students(roll, name, class), Subjects(sub\_id, sub\_name, max\_marks), Marks(roll, sub\_id, marks)
- Inserts at least 8 student records with marks in 4 subjects
- Performs the following operations:
  1. Display all students with their marks
  2. Calculate and display each student's total and percentage
  3. Find the class topper
  4. Find subject-wise toppers
  5. List students who failed (percentage < 40%)
  6. Display the rank list
- Uses proper exception handling for database operations
- Closes the connection properly

### Task 3: Python - Data Analysis with CSV (5 marks)

Write a Python program that:

- Reads a CSV file sales\_data.csv with columns: Date, Product, Quantity, Price, Region

- Performs the following analyses:
  1. Total sales amount by region
  2. Best-selling product (by quantity)
  3. Daily sales trend (group by date)
  4. Region with highest average order value
  5. Products with zero or negative stock
- Writes the analysis report to a new file `analysis_report.txt`
- Uses `try-except` for file operations
- Displays results in a formatted table

## Viva Voce (5 marks)

1. What is a **primary key**? Can a table have composite primary key? Give an example. (1 mark)
2. Explain the difference between `INNER JOIN` and `LEFT JOIN` with diagrams. When would you use each? (1 mark)
3. What is the purpose of `GROUP BY` clause? How is it different from `ORDER BY`? (1 mark)
4. How does Python's `sqlite3` module handle database connections? What happens if you forget to `commit()` changes? (1 mark)
5. What is **normalization** in databases? Explain 1NF and 2NF with an example. (1 mark)

## Part D: Computational Thinking Skills (Internal Assessment)

**Instructions:** Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

| CT Skill                   | Emerging (1)                                               | Developing (2)                                           | Proficient (3)                                                       |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
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**Student Name: *Date:***

**Total CT Score: / 18 *Teacher Remarks:***

## **Part E: Computing Journal (Portfolio)**

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- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

# ICT Class 10 - Summative Assessment 1

Summative Assessment (SA-1)

- Class 10

Class: Class 10

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Networking Concepts, Internet and Web Technologies, HTML Fundamentals, CSS Styling

Curated and developed by

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4. Marks are indicated against each question.

## Part A: MCQ (10 x 1 = 10 marks)

- a. Which layer of the OSI model is responsible for end-to-end communication?
- b. Physical layer
- c. Data link layer
- d. Network layer
- e. Transport layer
- b. Which of the following is a Class C IP address?
- c. 10.0.0.1
- d. 172.16.0.1
- e. 192.168.1.1
- f. 224.0.0.1
- c. The `<!DOCTYPE html>` declaration in HTML5:
- d. Creates a paragraph
- e. Tells the browser to render in quirks mode
- f. Tells the browser this is an HTML5 document
- g. Is optional and can be removed
- d. Which CSS property creates space between the element border and its content?
- e. margin
- f. padding
- g. border-spacing
- h. gap
- e. Which HTML5 semantic tag represents a self-contained composition?
- f. `<div>`
- g. `<section>`
- h. `<article>`
- i. `<aside>`
- f. What is the purpose of `meta charset="UTF-8"` in HTML?
- g. Sets the page title
- h. Defines character encoding

- i. Adds viewport settings
- j. Links external stylesheet
- g. Which HTTP status code indicates “Page Not Found”?
  - h. 200
  - i. 301
  - j. 404
  - k. 500
- h. In CSS, which selector has the **highest** specificity?
  - i. Element selector p
  - j. Class selector .box
  - k. ID selector #main
  - l. Universal selector \*
- i. Which tag is used to embed a video in HTML5?
  - j. <video>
  - k. <embed>
  - l. <media>
  - m. <movie>
- j. The position: sticky CSS property:
  - k. Positions element relative to the viewport
  - l. Positions element relative to its normal position until scroll threshold
- m. Positions element relative to parent
- n. Removes element from document flow

## Part B: Short Answer (8 x 2 = 16 marks)

- a. Explain the difference between **IPv4** and **IPv6**. Why was IPv6 introduced? Give the format of each.
- b. Write the HTML code for a form that collects: Full Name, Email, Phone (with pattern validation), Date of Birth, Gender (radio buttons), and a Submit button. Include proper labels and validation attributes.
- c. Differentiate between **block-level** and **inline** elements with three examples each. Which elements are inline-block?
- d. What is the 盒模型 (box model) in CSS? Explain content, padding, border, and margin with a labelled diagram.
- e. Explain the TCP/IP model with its four layers. Name two protocols at each layer.
- f. Write CSS rules to:
  - Create a responsive navigation bar with flexbox
  - Style links with hover underline effect
  - Set a media query for screens below 768px width
  - Create a 3-column grid layout

- g. What is **responsive web design**? Explain three techniques used to achieve it (media queries, flexbox, relative units).
- h. Differentiate between `class` and `id` selectors in CSS. Explain specificity rules with examples.

## Part C: Long Answer (4 x 3 = 12 marks)

- a. Create a complete HTML page for a **Student Management System** that includes:
- A header with school name and logo placeholder
  - A form to add new students (name, roll, class, section, marks)
  - A styled table displaying 5 students with sorting capability
  - CSS for professional formatting with borders, colours, spacing
  - Footer with contact information

### Previous Year Pattern - Karnataka Board 2024

- b. Write a detailed comparison of **Bus**, **Star**, and **Mesh** topologies. For each, explain:
- Structure with a diagram
  - How data flows
  - Two advantages and two disadvantages
  - Best use case scenario

### Board Exam Pattern - Karnataka 2023

- c. Create a complete CSS-styled webpage for a **Restaurant Menu** that includes:
- Hero section with restaurant name and tagline
  - Menu section with food items in categories (Starters, Main Course, Desserts)
  - Use CSS Grid for layout
  - Responsive design that works on mobile
  - Proper typography and colour scheme
- d. Explain **DNS resolution process** step by step. Include:
- What is DNS and why it is needed
  - Types of DNS servers (recursive, root, TLD, authoritative)
  - The complete resolution process with a diagram
  - What happens when you type `www.example.com` in the browser

## Part D: Application Based (2 x 1 = 2 marks)

- a. A student creates a webpage where the navigation bar disappears on mobile. The CSS code is:

```
nav {
 display: flex;
 justify-content: space-around;
}
@media (max-width: 768px) {
 nav {
 display: none;
 }
}
```

Identify the problem and write the corrected CSS that shows a hamburger menu on mobile. Explain your approach.

b. A company has three offices in Bangalore, Mysore, and Hubli. They want to:

- Connect all offices with high-speed data transfer
- Share a common database server
- Allow video conferencing between offices

Recommend the appropriate network type (LAN/MAN/WAN) and explain why. Suggest two protocols that would be used.

## Part D: Computational Thinking Skills (Internal Assessment)

**Instructions:** Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

| CT Skill                   | Emerging (1)                                               | Developing (2)                                           | Proficient (3)                                                       |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| <b>Decomposition</b>       | Needs help breaking down complex tasks into smaller parts  | Can break down tasks with some guidance or prompts       | Independently breaks down complex problems into manageable sub-tasks |
| <b>Pattern Recognition</b> | Cannot identify similarities or trends in data or problems | Identifies patterns with prompts or scaffolding          | Independently identifies and applies patterns to solve new problems  |
| <b>Abstraction</b>         | Includes all details, cannot filter irrelevant information | Removes some irrelevant details with guidance            | Filters to essential information, focuses on what matters            |
| <b>Algorithm Design</b>    | Provides random or disordered steps                        | Creates sequential but incomplete step-by-step solutions | Designs clear, ordered, complete algorithms with start and end       |
| <b>Debugging</b>           | Gives up when errors occur                                 | Finds and fixes errors with help from teacher or peers   | Systematically tests, identifies, and fixes errors independently     |
| <b>Evaluation</b>          | Does not check if solution works                           | Tests solution with guidance                             | Evaluates efficiency and correctness of own and others' solutions    |

**Student Name:** *Date:*

**Total CT Score:** / 18 **Teacher Remarks:**

## Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

# ICT Class 10 - Summative Assessment 2

Summative Assessment (SA-2)

- Class 10

Class: Class 10

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Ch 1-6: Networking, HTML/CSS, Python Programming, SQL Database Management

Curated and developed by

**Hoysala T, Computer Science Teacher**

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## General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

## Part A: MCQ (10 x 1 = 10 marks)

a. What will be the output?

```
x = [1, 2, 3]
y = x
y.append(4)
print(x)
```

- a. [1, 2, 3]
  - b. [1, 2, 3, 4]
  - c. [4, 1, 2, 3]
  - d. Error
- b. Which SQL command is used to add a new column to an existing table?
- c. ADD COLUMN
  - d. ALTER TABLE ... ADD
  - e. INSERT COLUMN
  - f. CREATE COLUMN
- c. Which CSS property is used to make text bold?
- d. font-weight: bold
  - e. text-style: bold
  - f. font-style: bold
  - g. text-weight: bold
- d. What does try-except block do in Python?
- e. Ignores all errors
  - f. Handles exceptions gracefully
  - g. Terminates the program
  - h. Creates error logs
- e. Which HTML tag creates a hyperlink?
- f. <link>
  - g. <a>
  - h. <href>
  - i. <url>

- f. Which SQL function returns the current date and time?
- g. GETDATE()
- h. NOW()
- i. CURRENT\_TIMESTAMP
- j. Both B and C
- g. What is the output?
- ```
s = "Hello World"
print(s.split()[1])
```
- a. Hello
- b. World
- c. Hello World
- d. Error
- h. Which CSS display value makes an element invisible but still takes space?
- i. display: none
- j. visibility: hidden
- k. opacity: 0
- l. display: invisible
- i. In Python, which keyword is used for exception re-raising?
- j. raise
- k. throw
- l. except
- m. catch
- j. Which SQL query displays the structure of a table?
- k. SHOW TABLE
- l. DESCRIBE table_name
- m. STRUCTURE table_name
- n. INFO table_name

Part B: Short Answer (8 x 2 = 16 marks)

- a. Write Python code to read a CSV file containing student records (Name, Roll, Marks) and display the top 3 scorers. Use the `csv` module.
- b. Write the output of the following and explain each line:

```
import json
data = {"students": [{"name": "A", "marks": 85}, {"name": "B", "marks": 92}]}
json_str = json.dumps(data, indent=2)
print(json_str)
parsed = json.loads(json_str)
print(parsed["students"][1]["name"])
```

- c. Write SQL queries for an **Online Shopping** database:
- Customers(cid, name, email, city)

- Products(pid, name, price, stock)
- Orders(oid, cid, pid, quantity, order_date)

(a) Find customers from 'Bangalore' (b) List products with stock less than 10 (c) Find total orders placed by each customer (d) Find the most ordered product

- d. Differentiate between list and tuple in Python with 5 points. When would you use each?
- e. Write CSS code to create a **pricing card** component with:
- Card with shadow and rounded corners
 - Header with gradient background
 - Price display in large font
 - Feature list with checkmark icons
 - Call-to-action button with hover effect
- f. Explain **normalization** in databases. What is the difference between 1NF, 2NF, and 3NF? Give an example table that violates each.
- g. Write a Python function `merge_sort(arr)` that implements the merge sort algorithm. Include proper comments and handle edge cases.
- h. What is the purpose of JOIN in SQL? Explain the difference between INNER JOIN, LEFT JOIN, and RIGHT JOIN with diagrams and examples.

Part C: Long Answer (4 x 3 = 12 marks)

- a. Write a complete Python program for a **Library Management System** that:
- Uses file handling to store book records (title, author, ISBN, available_copies)
 - Implements functions for: add_book, search_book, issue_book, return_book, display_all
 - Handles exceptions for file operations and invalid inputs
 - Generates a daily report of issued and returned books

Board Exam Pattern - Karnataka 2024

- b. Create a complete HTML/CSS webpage for a **Healthcare Appointment System** that includes:
- Header with hospital name and navigation
 - Appointment booking form (doctor selection, date, time, patient details)
 - Styled table showing available doctors with specialisation and timings
 - Responsive design with media queries
 - Professional medical-themed colour scheme

Previous Year Pattern - Karnataka Board 2023

- c. Write SQL queries for a **School Examination System** with tables:
- Students(roll, name, class, section)
 - Subjects(sub_id, sub_name, max_marks)
 - Marks(roll, sub_id, marks_obtained)
- (a) Create a **result card** showing each student's marks, total, and percentage (b) Find the class topper (c) Find subject-wise toppers (d) List students who failed in any subject (marks <

35% of max_marks) (e) Find the overall pass percentage (f) Display the rank list with rank, name, total, and percentage

d. Explain the complete process of **data communication** from one computer to another.

Include:

- Components of data communication (sender, receiver, medium, protocol)
- Types of transmission (simplex, duplex, multiplex)
- Errors in data transmission and how they are detected (parity check, CRC)
- Role of TCP in reliable data transfer with a diagram

Part D: Application Based (2 x 1 = 2 marks)

a. A student writes the following Python code to read a file:

```
f = open("data.txt")
content = f.read()
print(content)
f.close()
```

The program crashes if the file doesn't exist. Identify the issue and rewrite the code using try-except and with statement. Also explain why with is preferred.

b. A database has the following Students table:

roll	name	class	section	marks
1	Rahul	10	A	85
2	Priya	10	A	92
3	Amit	10	B	78

Write the SQL query that produces this output:

section	avg_marks	topper
A	88.5	Priya
B	78.0	Amit

Explain each clause of your query.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems

Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 10 - Unit Test 1

Unit Test 1

- Class 10

Class: Class 10

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Networking Concepts

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

- a. Which network topology requires a central hub or switch?
 - b. Bus topology
 - c. Star topology
 - d. Ring topology
 - e. Mesh topology
- b. The OSI model has how many layers?
 - c. 5
 - d. 6
 - e. 7
 - f. 8
- c. Which of the following is a Class B IP address?
 - d. 192.168.1.1
 - e. 172.16.0.1
 - f. 10.0.0.1
 - g. 127.0.0.1
- d. Which protocol is used to send email?
 - e. HTTP
 - f. FTP
 - g. SMTP
 - h. POP3
- e. What is the purpose of a firewall?
 - f. To boost internet speed
 - g. To filter incoming and outgoing network traffic
 - h. To store backup data
 - i. To manage user accounts

Section II: Short Answer (5 x 1 = 5 marks)

- a. Explain the difference between **IPv4** and **IPv6** addressing. Why is IPv6 needed?

- Draw a neat labelled diagram of **Star Topology**. List two advantages and two disadvantages.
- Write the full forms and explain the purpose of: HTTP, FTP, SMTP, DNS, TCP.
- What is a **MAC address**? How is it different from an **IP address**?
- Explain how data travels from one computer to another through the Internet, mentioning at least 3 devices/protocols involved.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
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Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 10 - Unit Test 2

Unit Test 2

- Class 10

Class: Class 10

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: HTML and CSS

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

- a. Which tag is used to define an internal style sheet in HTML?
 - b. <css>
 - c. <style>
 - d. <script>
 - e. <link>
- b. Which HTML5 element is used for navigation?
 - c. <nav>
 - d. <navigation>
 - e. <navigate>
 - f. <menu>
- c. What is the default display property of a <div> element?
 - d. inline
 - e. block
 - f. flex
 - g. grid
- d. Which CSS unit is relative to the font-size of the element?
 - e. px
 - f. em
 - g. cm
 - h. pt
- e. Which pseudo-class is used to style an element when the mouse hovers over it?
 - f. :active
 - g. :focus
 - h. :hover
 - i. :visited

Section II: Short Answer (5 x 1 = 5 marks)

- Write the HTML code to create a form with fields: Username, Password, Email, and a Submit button. Include proper labels and placeholder attributes.
- Differentiate between **relative positioning** and **absolute positioning** in CSS. Give one use case for each.
- What is the purpose of z-index property? Under what condition does it work?
- Write CSS code to create a 3-column responsive grid layout using CSS Grid that collapses to 1 column on mobile.
- Explain the difference between `margin: auto` and `text-align: center`. When would you use each?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)

- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 10 - Unit Test 3

Unit Test 3

- Class 10

Class: Class 10

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: Python Programming

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. What will be the output?

```
s = "PYTHON"
print(s[1:4])
```

- a. PYT
 - b. PYTH
 - c. YTH
 - d. YTH0
- b. Which of the following is used to create a dictionary?
- c. {}
 - d. []
 - e. ()
 - f. <>
- c. What does `strip()` method do in Python?
- d. Removes all spaces
 - e. Removes leading and trailing whitespace
 - f. Removes all characters
 - g. Reverses the string
- d. Which module is used to work with regular expressions?
- e. regex
 - f. re
 - g. regexp
 - h. pyre
- e. What will be the output?
- ```
print([1, 2, 3] * 2)
```
- a. [1, 2, 3, 1, 2, 3]
  - b. [[1, 2, 3], [1, 2, 3]]
  - c. Error
  - d. [2, 4, 6]

## Section II: Short Answer (5 x 1 = 5 marks)

a. Write the output and explain:

```
fruits = ["apple", "banana", "cherry"]
for i, fruit in enumerate(fruits):
 print(f"{i+1}. {fruit}")
```

b. Differentiate between `split()` and `join()` string methods with one example each.

c. Write a Python function `is_palindrome(s)` that checks if a string is a palindrome (case-insensitive). Test it with 3 examples.

d. Explain the difference between a **list comprehension** and a **generator expression**. Give one example of each.

e. Write the output of:

```
d = {"a": 1, "b": 2, "c": 3}
print(list(d.keys()))
print(list(d.values()))
print(d.get("d", 0))
```

## Part D: Computational Thinking Skills (Internal Assessment)

**Instructions:** Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

| CT Skill                   | Emerging (1)                                               | Developing (2)                                           | Proficient (3)                                                       |
|----------------------------|------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| <b>Decomposition</b>       | Needs help breaking down complex tasks into smaller parts  | Can break down tasks with some guidance or prompts       | Independently breaks down complex problems into manageable sub-tasks |
| <b>Pattern Recognition</b> | Cannot identify similarities or trends in data or problems | Identifies patterns with prompts or scaffolding          | Independently identifies and applies patterns to solve new problems  |
| <b>Abstraction</b>         | Includes all details, cannot filter irrelevant information | Removes some irrelevant details with guidance            | Filters to essential information, focuses on what matters            |
| <b>Algorithm Design</b>    | Provides random or disordered steps                        | Creates sequential but incomplete step-by-step solutions | Designs clear, ordered, complete algorithms with start and end       |
| <b>Debugging</b>           | Gives up when errors occur                                 | Finds and fixes errors with help from teacher or peers   | Systematically tests, identifies, and fixes errors independently     |
| <b>Evaluation</b>          | Does not check if solution works                           | Tests solution with guidance                             | Evaluates efficiency and correctness of own and others' solutions    |

**Student Name:** *Date:*

**Total CT Score:** / 18 **Teacher Remarks:**

## Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

# ICT Class 10 - Unit Test 4

## Unit Test 4

- Class 10

Class: Class 10

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL Database Management

Curated and developed by

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## General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

## Section I: MCQ (5 x 1 = 5 marks)

- a. Which SQL clause is used to sort the result set?
  - b. SORT BY
  - c. ORDER BY
  - d. GROUP BY
  - e. ARRANGE BY
- b. Which of the following is a DDL command?
  - c. SELECT
  - d. INSERT
  - e. CREATE
  - f. UPDATE
- c. What is the correct order of SQL clauses in a SELECT statement?
  - d. SELECT - FROM - WHERE - GROUP BY - ORDER BY
  - e. SELECT - WHERE - FROM - GROUP BY - ORDER BY
  - f. FROM - SELECT - WHERE - GROUP BY - ORDER BY
  - g. ORDER BY - SELECT - FROM - WHERE
- d. Which function returns the number of rows in a table?
  - e. SUM()
  - f. COUNT()
  - g. TOTAL()
  - h. ROWS()
- e. Which constraint ensures all values in a column are unique?
  - f. PRIMARY KEY
  - g. UNIQUE
  - h. NOT NULL
  - i. CHECK

## Section II: Short Answer (5 x 1 = 5 marks)

- a. Write SQL queries for a table Products(pid, name, category, price, stock):

- Find products with price between 100 and 500
  - Count products in each category
  - Find the product with maximum price
  - Display products sorted by stock (descending)
- b. What is a **foreign key**? How does it establish a relationship between two tables? Give an example.
- c. Write the output of:
- ```
SELECT ROUND(45.678, 2);
SELECT MOD(17, 5);
SELECT NOW();
```
- d. Differentiate between DELETE and DROP commands.
- e. Explain JOIN with an example. Write a query to display student names and their marks from two tables Students and Marks.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems
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Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: *Date:*

Total CT Score: / 18 **Teacher Remarks:**

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)